

Article

Lithium-Decorated C₂₆ Fullerene in DFT Investigation: Tuning Electronic Structures for Enhanced Hydrogen Storage

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Abstract

Hydrogen energy holds immense potential to address the global energy crisis and environmental challenges. However, its large-scale application is severely hindered by the lack of efficient hydrogen storage materials. This study systematically investigates the H₂ adsorption properties of intrinsic C₂₆ fullerene and Li-decorated C₂₆ fullerene using density functional theory (DFT) calculations. The results reveal that Li atoms preferentially bind to the H₅₋₅ site of C₂₆, driven by significant electron transfer (0.90 |e|) from Li to C₂₆. This electron redistribution modulates the electronic structure of C₂₆, as evidenced by projected density of states (PDOS) analysis, where the p orbitals of C atoms near the Fermi level undergo hybridization with Li orbitals, enhancing the electrostatic environment for H₂ adsorption. For Li-decorated C₂₆, the average adsorption energy and consecutive adsorption energy decrease as more H₂ molecules are adsorbed, indicating a gradual weakening of adsorption strength and signifying a saturation limit of three H₂ molecules. Charge density difference and PDOS analyses further demonstrate that H₂ adsorption induces synergistic electron transfer from both Li (0.89 |e| loss) and H₂ (0.01 |e| loss) to C₂₆ (0.90 |e| gain), with orbital hybridization between H s orbitals, C p orbitals, and Li orbitals stabilizing the adsorbed system. This study aimed to provide a comprehensive understanding of the microscopic mechanism underlying Li-enhanced H₂ adsorption on C₂₆ fullerene and offer insights into the rational design of metal-decorated fullerene-based systems for efficient hydrogen storage.



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Keywords: density functional theory; H₂ adsorption; C₂₆ fullerene; electron transfer

1. Introduction

As the global energy demand soars incessantly and the alarm over environmental challenges intensifies, the quest for energy storage solutions that are both efficacious and enduring has ascended to a critical imperative [1,2]. Hydrogen, endowed with high energy density, renewability, and friendliness toward the environment, emerges as a promising clean energy carrier, making it a potential substitute for fossil fuels in the future energy landscape [3,4]. Yet, the large-scale application of hydrogen energy is limited by the difficulties in its storage and transportation, particularly the lack of efficient and reversible hydrogen storage materials at room temperature [5,6]. The efficient storage of hydrogen is a critical challenge that must be addressed to facilitate the widespread adoption of hydrogen as an energy carrier [7].

Among various hydrogen storage methods, physical adsorption on solid materials has emerged as a particularly attractive option, offering the potential for high storage capacity, rapid adsorption kinetics, and low energy consumption [8,9]. In recent years, carbon-based nanomaterials, such as carbon nanotubes, graphene, and fullerenes, have garnered significant interest for hydrogen storage because of their unique structures and properties [10,11]. Fullerenes with their hollow and symmetrical cage-like structures present a sufficient space for the physisorption of hydrogen molecules [12,13]. Compared to common fullerenes such as C₆₀, the larger curvature of C₂₆ may result in a more specific electron cloud distribution, thereby influencing its interaction patterns with other substances [14,15]. Meanwhile, the smaller molecular size could endow it with unique advantages in certain application scenarios, such as the fabrication of nanoscale devices or targeted delivery in the biomedical field [16,17]. Owing to its unique structural and electronic properties, C₂₆ fullerene is expected to serve as a potential high-efficiency hydrogen storage material [18].

The storage capacity of pristine fullerenes, however, is limited by their weak van der Waals interactions with hydrogen molecules [19]. To enhance the hydrogen storage performance of fullerenes, various strategies have been explored, including the modification of fullerenes with metal atoms [20,21]. Metal decoration has been identified as an effective means to strengthen the interaction between fullerenes and hydrogen molecules, thereby increasing the storage capacity and reversibility of the system [22,23].

Lithium (Li), an alkali metal with a high affinity for hydrogen, stands out as a potential modifier for modifying fullerenes to enhance hydrogen storage [24]. The interaction between Li and hydrogen is strong enough to provide a significant binding energy, while the low weight of Li contributes to a high gravimetric storage capacity [25]. Moreover, the electronic structure of Li-decorated fullerenes can be tuned to optimize the hydrogen adsorption and desorption processes under mild conditions, which is essential for practical applications [26].

Therefore, it is necessary to investigate the H₂ adsorption properties of C₂₆ fullerene to understand its hydrogen storage mechanism. In this study, the adsorption sites, interaction energies, and electronic structures of H₂ adsorbed on intrinsic C₂₆ fullerene and Li-modified C₂₆ fullerene were studied using density functional theory (DFT) methods to evaluate their potential as hydrogen storage media. Our work aims to contribute to the development of efficient hydrogen storage materials, which are crucial for the practical implementation of hydrogen fuel cell technologies and the transition to a sustainable energy economy.

2. Results and Discussion

2.1. Hydrogen Adsorption on C₂₆ Fullerene

From the top view and side view of the C₂₆ fullerene in Figure 1, we can gain insights into its structural characteristics, such as atomic arrangement and symmetry. In terms of the atomic arrangement, the C₂₆ molecule consists of 26 carbon atoms, which are interconnected via covalent bonds to form a closed cage-like structure with D_{3h} symmetry. The surface of the C₂₆ molecule contains three hexagonal rings and twelve pentagonal rings.

To further identify the most stable adsorption site of H₂ on C₂₆ fullerene, the different sites were considered to perform energy calculations. It was found that H₂ exhibits the lowest adsorption energy (−0.25 eV) at the H₅₋₅ adsorption site, indicating that this is the most stable adsorption site for H₂. Upon further adsorption of hydrogen, the calculated stepwise adsorption energy is −0.01 eV, which is close to zero. This indicates that C₂₆ fullerene can only adsorb one hydrogen molecule, implying that the intrinsic C₂₆ fullerene is not suitable as a hydrogen carrier.

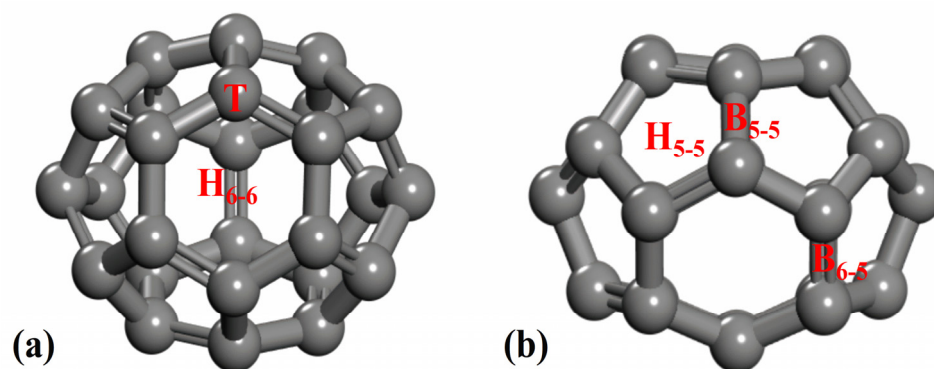


Figure 1. The (a) top and (b) side views of the C_{26} fullerene structure. The C atoms are shown in grey.

2.2. Li-Decorated C_{26} Fullerene

The binding characteristics of Li atoms on different sites of C_{26} fullerene were investigated, as presented in Table 1. The results reveal that the H_{5-5} site exhibits the highest binding energy (-2.68 eV) among all considered positions, indicating the most thermodynamically favorable adsorption configuration for Li decoration. This strong binding is accompanied by a relatively short C-Li distance of 2.22 Å. In contrast, the Li atom bonded at the T site is dynamically unstable, prompting Li atoms to migrate spontaneously to the H_{5-5} site, highlighting the intrinsic preference for this specific adsorption geometry. The H_{6-6} and B_{5-5} sites also demonstrate substantial binding energies (-2.59 eV and -2.42 eV, respectively), with corresponding distances of 2.28 Å and 2.11 Å, reflecting moderate interaction strengths. The B_{6-5} site, while still stable, exhibits the lowest binding energy (-2.38 eV) among the viable configurations, which is attributed to its intermediate coordination environment.

Table 1. The binding energy (E_b) of the Li atom on C_{26} fullerene and the distance (d_{C-Li} , Å) between Li and C_{26} at different sites of Li-decorated C_{26} fullerene.

Site	E_b (eV)	d_{C-Li} (Å)
H_{5-5}	-2.68	2.22
H_{6-6}	-2.59	2.28
B_{5-5}	-2.42	2.11
B_{6-5}	-2.38	2.12
T	unstable and move to H_{5-5} site	

The cohesive energy of the solid bulk phase of the Li atom is 1.13 eV/atom [27], which is less than the binding energy of the Li atom on C_{26} fullerene, suggesting that the Li atom tends to bind to C_{26} fullerene rather than aggregating with each other. These findings suggest that Li-decorated C_{26} is most effective at the H_{5-5} site, which may significantly modulate the electronic structure of the C_{26} fullerene, thereby enhancing its potential as a hydrogen storage medium.

2.3. Interaction Between Li and C_{26} Fullerene

To further explore the intrinsic relationship between the electronic structure of Li and C_{26} fullerene, the charge density difference for the Li-decorated C_{26} fullerene was studied. As shown in Figure 2b, a region of electron depletion (red) is observed around the Li atom, with a region of electron accumulation (blue) around the C_{26} fullerene. This indicates that electron transfer occurs from the Co surface to the C atom. Bader charge analysis [28,29] showed that 0.90 electrons ($0.90 |e|$) are transferred from Li atoms to C_{26} fullerene after the interaction between them. This charge transfer process is attributed to the electronegativity

difference between Li atoms and C_{26} fullerene. Li atoms exhibited low electronegativity and weak electron-binding ability, whereas C_{26} fullerene had relatively higher electronegativity and stronger electron-attracting capacity. The increased electron cloud density of C_{26} fullerene after gaining electrons may enhance its ability to adsorb hydrogen.

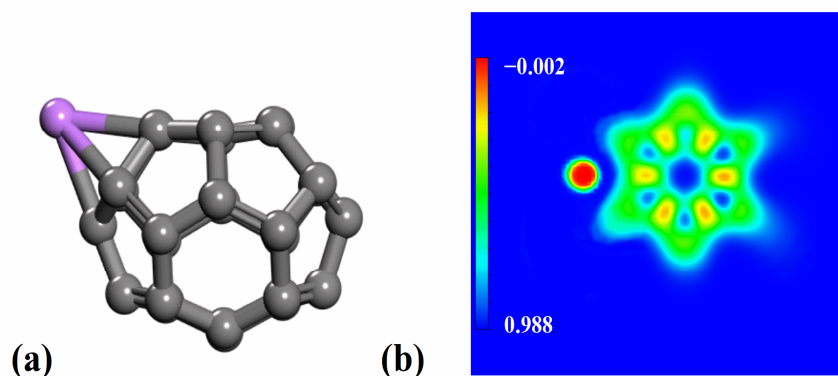


Figure 2. (a) Optimized structure and (b) charge density difference of Li–decorated C_{26} fullerene. The C atoms and Li atom are shown in grey and purple, respectively. Red regions indicate charge depletion, while green and blue represent regions of reduced and increased charge density, respectively. The isosurfaces are set to $0.16 \text{ e}\text{\AA}^{-3}$.

To gain an in-depth understanding of the interaction between Li and C_{26} fullerene, a detailed analysis was performed on the projected density of states (PDOS) of C_{26} fullerene with Li decoration. As observed in Figure 3a, the density of states C_{26} molecule is mainly composed of the s and p orbitals of the C atoms. Among these, a distinct distribution of the density of states of the p orbitals near the Fermi level was observed. This is due to the formation of a conjugated system via covalent bonding between carbon atoms in the C_{26} molecule, which allows p-orbital electrons to be delocalized within the molecule, thereby influencing the electronic properties of C_{26} fullerene.

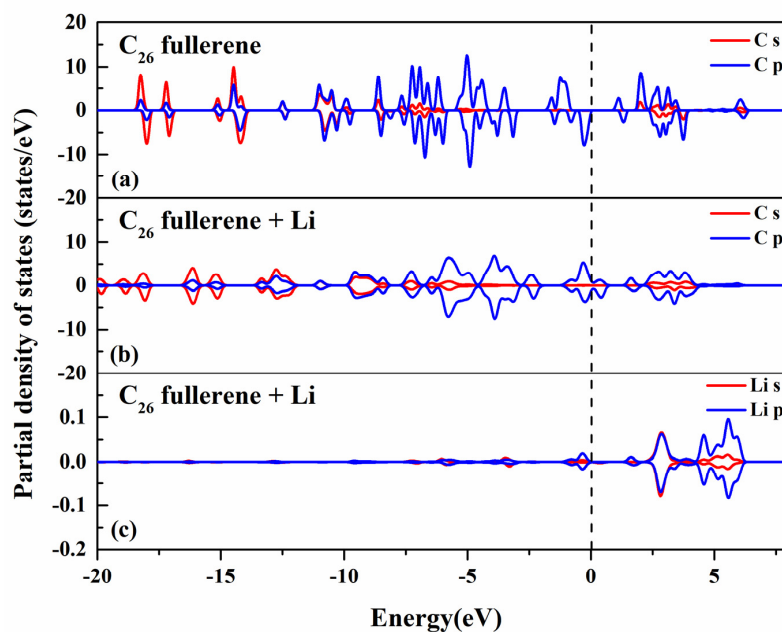


Figure 3. The projected density of states (PDOS) onto (a) s and p orbitals of the C atom of C_{26} fullerene; (b) s and p orbitals of the C atom of Li–decorated C_{26} fullerene; (c) s and p orbitals of the Li atom of Li–decorated C_{26} fullerene. The Fermi energy is set to zero. The s and p orbitals are labeled with red and blue lines, respectively.

After Li decoration, as shown in Figure 3b, the s orbital of C atoms becomes broader and more dispersed compared to pristine C₂₆ fullerene. The C s orbital exhibits broadened and less intense peaks compared to pristine C₂₆ fullerene, indicating electron redistribution induced by Li decoration. This effect is attributed to charge transfer from Li to C₂₆ fullerene, altering the electrostatic environment around C atoms. Near the Fermi level, the C p orbital shows reduced peak intensities and increased peak widths, suggesting weakened electron delocalization. This reduction is caused by electron injection from Li, which modifies the hybridization state of C atoms. From Figure 3c, it can be seen that the s and p orbitals of the Li atom show sparse low-intensity peaks concentrated at higher energies (0 eV to 5 eV), consistent with the alkali-metal character of Li.

The PDOS analysis demonstrates that Li decoration modifies the electronic structure of C₂₆ through electron transfer and weak orbital hybridization. These changes enhance the electron density on C₂₆, particularly around the Fermi level, which may improve its ability to adsorb hydrogen by providing favorable electrostatic interactions and increased orbital overlap with H₂ molecules.

2.4. Hydrogen Adsorption on Li-Decorated C₂₆ Fullerene

To investigate the role of Li in enhancing the hydrogen adsorption performance of C₂₆ fullerene, the optimized structures of Li-decorated C₂₆ fullerene with different numbers of adsorbed nH₂ molecules were investigated, as depicted in Figure 4. For the structure with one H₂ adsorption (Figure 4a), the single H₂ molecule is positioned near the Li-C₂₆ with the average adsorption energy of −0.15 eV. When two H₂ molecules are adsorbed (Figure 4c), the second H₂ molecule attaches to the system in a distinct orientation relative to the first one, suggesting that the surface of the Li-decorated C₂₆ has additional available sites to accommodate more H₂. As for the structure with three H₂ adsorptions (Figure 4d), the third H₂ molecule and the first two H₂ molecules present a triangular symmetric position.

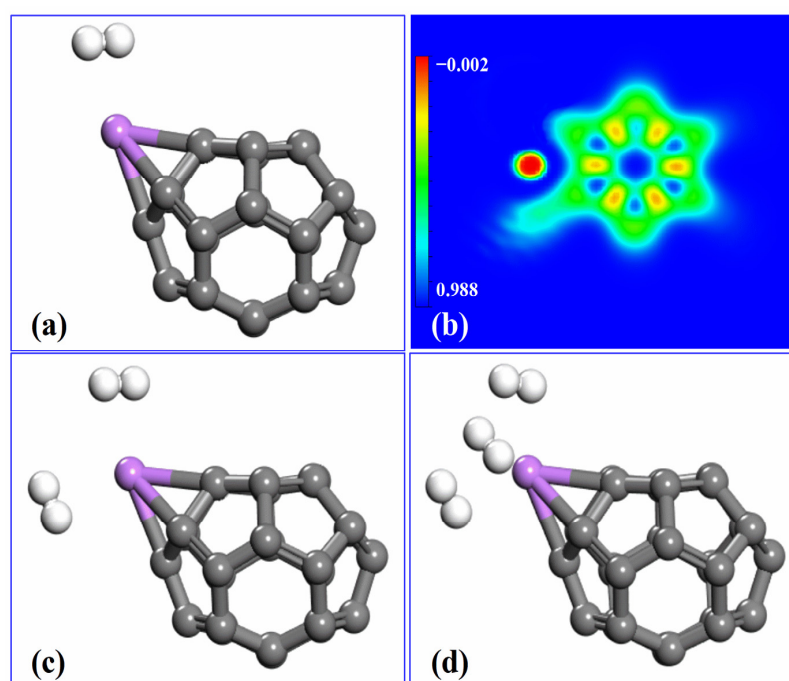


Figure 4. The optimized structure of Li-decorated C₂₆ fullerene with (a) 1 H₂ adsorption, (c) 2 H₂ adsorption, and (d) 3 H₂ adsorption. (b) Charge density difference of Li-decorated C₂₆ fullerene with H₂ adsorption. The H, C, and Li atoms are shown in white, grey, and purple, respectively. Red regions indicate charge depletion, while green and blue represent regions of reduced and increased charge density, respectively. The isosurfaces are set to 0.16 eÅ^{−3}.

The average adsorption energy, consecutive adsorption energy, and Li-H₂ distance for increasing numbers of H₂ molecules adsorbed onto Li-decorated C₂₆ fullerene are shown in Table 2. As the number of adsorbed H₂ molecules (n) increases, the average adsorption energy shows a gradual decreasing trend, dropping from −0.15 eV when n = 1 to −0.09 eV when n = 4. This indicates that the adsorption capacity of C₂₆ for H₂ is gradually weakened.

Table 2. The average adsorption energy (E_{ads}), consecutive adsorption energy (E_c) of the nH₂/Li-C₂₆ system, and corresponding distance ($d_{\text{C-Li}}$, Å) between Li and H₂.

$n\text{H}_2$	E_{ads} (eV)	E_c (eV)	$d_{\text{Li-H}_2}$ (Å)
1H ₂	−0.15	−0.15	2.09
2H ₂	−0.13	−0.11	2.13
3H ₂	−0.11	−0.07	2.25
4H ₂	−0.09	−0.02	2.44

The consecutive adsorption energy reflects the energy change during the adsorption process when one more hydrogen molecule is added. It can be observed from Table 2 that the consecutive adsorption energy gradually decreases, falling from −0.15 eV when n = 1 to −0.02 eV when n = 4. This means that as the number of already adsorbed hydrogen molecules increases, it becomes increasingly difficult to adsorb one more hydrogen molecule. When n reaches 4, the value of consecutive adsorption energy is −0.02 eV, close to zero, indicating that the maximum saturation number of hydrogen molecules adsorbed by the C₂₆ structure is 3, which is more than one H₂ adsorption of intrinsic C₂₆ fullerene. The distance increases from 2.09 Å (1H₂) to 2.44 Å (4H₂), implying a weakening interaction between Li and H₂ as more H₂ molecules are adsorbed.

2.5. Interaction of Hydrogen with Li-Decorated C₂₆ Fullerene

To elucidate the electronic effects of H₂ adsorption, the charge density difference of H₂ adsorption on Li-decorated C₂₆ fullerene was studied, as shown in Figure 4b. The electron transfer was quantified by Bader charge analysis: after hydrogen absorption, Li loses 0.89 |e|, H₂ loses 0.01 |e|, and C₂₆ gains 0.90 |e|. Upon H₂ adsorption, the charge depletion around Li becomes more pronounced in the H₂ adsorbed system (broader red-green regions), indicating a strengthened electron-donating role of Li under H₂ interaction. Conversely, the charge accumulation on C₂₆ expands and intensifies (more extensive blue regions) compared to the Li-decorated C₂₆ system, which is attributed to the combined electron transfer from both Li and H₂. The decoration of Li to C₂₆ leads to the spatial charge redistribution, which enhances H₂ adsorption stability to its saturation at three H₂ molecules, consistent with the trends in adsorption energy, providing a microscopic storage mechanism for C₂₆ fullerene.

The PDOS of Li-decorated C₂₆ fullerene after H₂ adsorption was calculated to explore the interaction of H₂ with Li-decorated C₂₆ fullerene, as shown in Figure 5. For the isolated H₂ molecule in Figure 5a, a sharp and intense peak is observed in the H s orbital at −17 eV. After H₂ adsorption onto Li-decorated C₂₆, as shown in Figure 5b, the peak of the H s orbital was shifted to a higher energy (−5 eV) and broadened. This change is attributed to electron transfer and orbital hybridization between H₂ and the C₂₆-Li system. The interaction disrupts the localized H₂ s orbital, redistributing electron density and altering the energy distribution of H electrons.

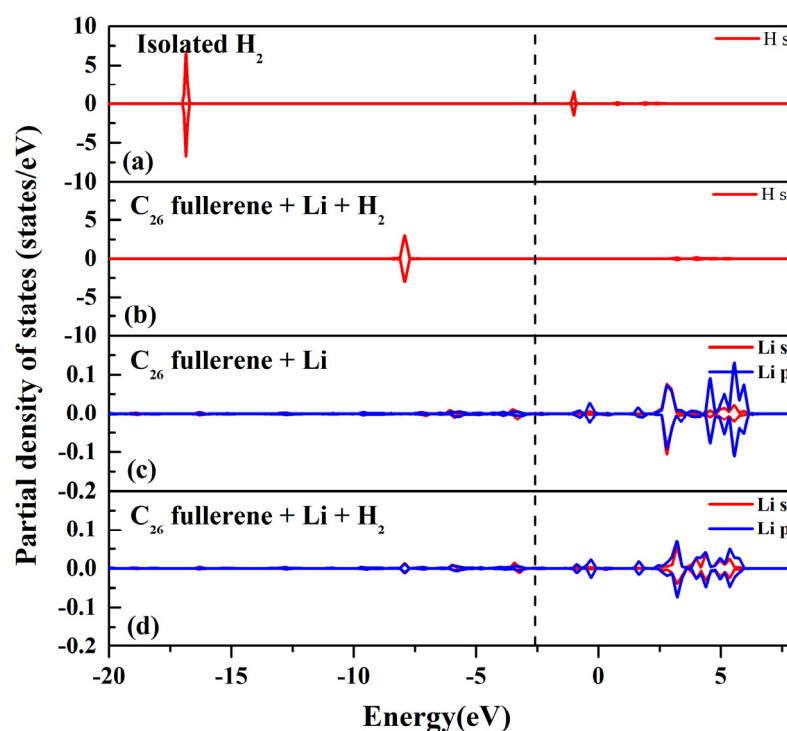


Figure 5. The projected density of states (PDOS) onto (a) s orbital of the H atom of isolated H_2 molecule; (b) s orbital of the H atom of Li-decorated C_{26} fullerene after H_2 adsorption; (c) s and p orbitals of the Li atom of Li-decorated C_{26} fullerene; (d) s and p orbitals of the Li atom of Li-decorated C_{26} fullerene after H_2 adsorption. The Fermi energy is set to zero. The s and p orbitals are labeled with red and blue lines, respectively.

As shown in Figure 5d, upon H_2 adsorption, the peak intensities in s and p orbitals of the Li atom were decreased, and distributions became more dispersed. This reduction in electron density is caused by electron transfer from Li to H_2 , modifying the local electronic environment of the Li atom. The synergistic effects of electron transfer and orbital hybridization resulted in preferential H_2 adsorption at Li-decorated C_{26} , exhibiting selectivity. This selective adsorption is of great significance for the development of efficient hydrogen storage materials, as it can improve both the storage capacity and selectivity of C_{26} fullerene.

2.6. Kubas-Type Interaction

In our calculations, the H-H bond length of H_2 molecules adsorbed on Li-decorated C_{26} (for one to four H_2 molecules) was consistently 0.77 Å, which is slightly elongated compared to the 0.75 Å bond length of isolated H_2 . This slight elongation (0.02 Å) aligns with the key characteristic of Kubas-type interactions, where H-H bond length changes moderately (neither drastic enough for chemisorption nor negligible for pure physisorption), as observed in the Y-decorated C_{24} system [30].

Additionally, our charge density difference and PDOS analyses reveal electron transfer from Li and H_2 to C_{26} , with orbital hybridization between H s orbitals, C p orbitals, and Li orbitals. This orbital interaction, coupled with the observed H-H bond elongation, suggests a mechanism analogous to the Kubas interaction described for Y-decorated systems, where charge transfer and weak orbital hybridization contribute to an interaction strength between physisorption and chemisorption.

Notably, the H-H bond length remains nearly constant as more H_2 molecules are adsorbed, while the distance between Li and H centers increases slightly. This stability in bond elongation further supports the presence of Kubas-type interactions, as the interaction

mechanism remains consistent across different H₂ coverage levels. These observations indicate that Kubas-type interactions play a significant role in H₂ adsorption on Li-decorated C₂₆, contributing to the enhanced adsorption capacity compared to intrinsic C₂₆ fullerene. In future research, we will further elaborate on the orbital-level charge transfer details to better quantify this interaction.

3. Computational Details

Computational Methods

Density functional theory (DFT) computations were employed by the Vienna ab initio simulation package (VASP) [31,32] 6.1.0. The interactions between electrons and ions were represented by the projector augmented wave (PAW) method [33,34]. The exchange-correlation functional was determined within the framework of the generalized gradient approximation (GGA) with Perdew–Burke–Ernzerhof (PBE) functional [35,36]. Spin polarization was calculated during the process of geometric optimization, and van der Waals (vdW) interactions were considered by the DFT-D3 method with Becke–Johnson damping [27]. A plane wave basis set cutoff energy [37] of 450 eV was applied. The convergence criteria of geometric optimization process for energy and force thresholds were set to 10^{−5} eV and 0.02 eV/Å, respectively. To separate the interactions between adjacent slabs, a 30 Å vacuum was established. The k-point integration was sampled using the Monkhorst–Pack scheme [38] with a (1 × 1 × 1) grid. The vacuum layer was set to 20 Å to avoid the interaction of periodically repeated slabs.

The binding energy (E_b) of the Li atom to C₂₆ was calculated as follows:

$$E_b = E_{\text{Li+C}_{26}} - E_{\text{C}_{26}} - E_{\text{Li}} \quad (1)$$

where $E_{\text{Li+C}_{26}}$, $E_{\text{C}_{26}}$, and E_{Li} are the total energies of Li-decorated C₂₆ fullerene, clean C₂₆ fullerene, and isolated Li atoms, respectively. Hence, a negative binding energy suggests that the metal atom can be attached to the C₂₆.

The average adsorption energy (E_{ad}) and consecutive adsorption energy (E_c) of the C₂₆ fullerene to the hydrogen molecule were calculated as follows:

$$E_{\text{ad}} = (E_{n\text{H}_2+\text{Li+C}_{26}} - E_{\text{Li+C}_{26}} - nE_{\text{H}_2})/n \quad (2)$$

$$E_c = E_{n\text{H}_2+\text{Li+C}_{26}} - E_{(n-1)\text{H}_2+\text{Li+C}_{26}} - E_{\text{H}_2} \quad (3)$$

where $E_{n\text{H}_2+\text{Li+C}_{26}}$, $E_{(n-1)\text{H}_2+\text{Li+C}_{26}}$, and E_{H_2} are the total energies of $n\text{H}_2$ and $(n - 1)\text{H}_2$ adsorbed on the Li-decorated C₂₆ fullerene and the total energy of an isolated hydrogen molecule, respectively; n is the number of adsorbed hydrogen molecules.

The larger absolute value of adsorption energy indicates a stronger adsorption strength. The larger the negative value, the stronger the adsorption capacity. The consecutive adsorption energy is the adsorption energy when the n th hydrogen molecule adsorbed onto C₂₆, which already has $(n - 1)$ hydrogen molecules adsorbed. The small value of consecutive adsorption energy implies that the adsorption process is not energetically favored, which is typically not desired for efficient hydrogen storage materials.

4. Conclusions

In summary, this study systematically investigates the structural, electronic, and H₂ adsorption properties of Li-modified C₂₆ fullerenes through DFT calculations, unraveling the critical role of Li decoration in modulating H₂ adsorption behavior at the atomic and electronic levels. Intrinsic C₂₆ fullerene exhibits limited hydrogen storage capacity, with a maximum adsorption of only one H₂ molecule due to weak van der Waals interaction. Li

decoration emerges as an effective strategy to overcome this limitation. The strong binding of Li atoms to the H₅₋₅ site of C₂₆ triggers substantial electron transfer (0.90 |e|) from Li to C₂₆. PDOS reveals that Li decoration reconstructs the s and p orbitals of C atoms, altering electron distribution near the Fermi level and strengthening interactions with H₂. For Li-decorated C₂₆, the average adsorption energy and consecutive adsorption energy both decrease with increasing H₂ coverage, with a saturation capacity of three H₂ molecules. This saturation is attributed to the weakening Li-H₂ interaction and reduced electron donation from Li, as observed in the charge density. Bader charge analysis further reveals that H₂ adsorption induces synergistic electron transfer from Li and H₂ to C₂₆, with orbital hybridization between H s orbitals, C p orbitals, and Li orbitals stabilizing the adsorbed system, as evidenced by PDOS shifts in H s orbitals. These findings demonstrate that Li decoration enhances the hydrogen storage capacity of C₂₆ by modulating its electronic structure through electron transfer and orbital hybridization, providing a theoretical basis for designing C₂₆-based hydrogen storage materials.

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Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author(s).

Conflicts of Interest: There are no conflicts of interest to declare.

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